Engineering Report

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June 18, 2026

To: AeroGrid CTO **From:** Data Engineer **Subject:** Turbine Telemetry Analysis and Cloud Architecture Proposal

Executive Summary

The report investigates the findings by the python script into which turbines are failing. This was analysed using a python script. From these finding, a system architecture was proposed and a method to optimise costs was suggested.

Findings

Running of the python script shows that turbines T-04 and T-07 are failing. This will either be because these turbines are exhibiting an average temperature greater than 85°C or if the vibration exceeds 15 mm/s.

Turbines Requiring Maintenance

T-04

T-07

Architecture Justification

The current AeroGrid architecture consists of turbine sensors followed by one server which leads to a database. This is fragile as it does not facilitate for the robust network required to accomodate a scalable turbine network. Specifically, if the system is overloaded, this can result in a loss of critical data.

Scalability

To fix the issue of scalability, the architecture consists of turbine sensors followed by Kafka/Kinesis, followed by Lambda and finally storage. This allows the system to scale dynamically as the load on the servers due to the changing turbine network changes.

Reliability

The system proposed above is more reliable because if the database crashes, the queue ensures that critical data is not lost. Processes resume as normal after system recovery.

Cost Optimisation

Hot and Cold storage is used to seperate data which is accessed frequently and less frequently respectively. This means that older data which is not accessed frequently can be stored via cheaper means to cut costs. Hot storage holds data which clients rely on more frequently and hence must be reliable and robust even under critical conditions. Hence, it uses more expensive storage technologies.